Band-Limitedness and Reality of Zeros of the Inverse-Phase Pullback of the Hardy Z-Function

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Abstract

Let Z be the Hardy function and ϑ the Riemann–Siegel phase. Set $\Theta(t) = \vartheta(t) + ct$ with $c > 0$ chosen so $\Theta' > 0$ on $[t_0, \infty)$, and define

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}, \qquad u \geq u_0.$$

We show: Y extends to an entire function of exponential type ≤ 1, its analytic signal Y_+ belongs to the Hermite–Biehler class (has no zeros in the upper half-plane), and consequently Y has only real zeros.

1 Setup

1.1 The analytic signal

The band-limit theorem (prior paper) establishes: the time-averaged autocovariance

$$R(\tau) = \lim_{T\to\infty} \frac{1}{\Theta(T)} \int_{u_0}^{\Theta(T)} Y(u+\tau)Y(u)\, du$$

is the Fourier transform of a positive Borel measure ρ with $\operatorname{supp}\rho \subseteq [-1, 1]$. By the Gelfand–Kolmogorov (Cramér) spectral representation,

$$Y(u) = \int_{-1}^{1} e^{i\mu u}\, d\xi(\mu), \qquad \mathbb{E}[d\xi(\mu)\overline{d\xi(\mu')}] = \delta(\mu - \mu')\, d\rho(\mu), \qquad d\xi(-\mu) = \overline{d\xi(\mu)}.$$

Definition 1 (Analytic signal)**.** $Y_+(z) := 2\int_{[0,1]} e^{i\mu z}\, d\xi(\mu)$ for $z \in \mathbb{C}$.

By dominated convergence, Y_+ is entire of exponential type ≤ 1. On $\mathbb{R}$, $Y_+ = Y + i\,\mathcal{H}[Y]$ where $\mathcal{H}$ is the Hilbert transform. Set $Y_+^*(z) := \overline{Y_+(\bar{z})}$ and $\tilde{Y}(z) := \frac{1}{2}(Y_+(z) + Y_+^*(z))$. Then $\tilde{Y}$ is entire, real on $\mathbb{R}$, and $\tilde{Y}|_{\mathbb{R}} = Y$; by the identity theorem, $\tilde{Y}$ is the unique entire extension of Y.

1.2 Reformulation target

Lemma 2 (HB reduction)**.** *If Y_+ has no zeros in the open upper half-plane $\mathbb{C}^+$, then all zeros of $\tilde{Y}$ are real.*

Proof. If Y_+ has no zeros in $\mathbb{C}^+$, then for $z \in \mathbb{C}^+$, $|Y_+(z)| > 0$ and the function Y_+^*/Y_+ is holomorphic in $\mathbb{C}^+$, bounded by 1 on $\mathbb{R}$ (where $|Y_+^*| = |Y_+|$), hence by the maximum principle $|Y_+^*(z)/Y_+(z)| \le 1$ for $z \in \mathbb{C}^+$. The inequality is strict except on a set of measure zero; for strictness we use the stronger Krein–Akhiezer condition established in Theorem 5 below, yielding $|Y_+^*(z)| < |Y_+(z)|$ for $z \in \mathbb{C}^+$, i.e., $Y_+ \in \mathrm{HB}$.

For the Hermite–Biehler theorem: if $(Y_+ + Y_+^*)(z_0) = 0$ with $z_0 \in \mathbb{C}^+$, then $Y_+(z_0) = -Y_+^*(z_0)$, so $|Y_+(z_0)| = |Y_+^*(z_0)|$, contradicting strict HB. Conjugate symmetry excludes $\mathbb{C}^-$. Hence $2\tilde{Y} = Y_+ + Y_+^*$ has only real zeros. □ □

2 de Branges space structure

2.1 Paley–Wiener and the de Branges space of E

Let PW_1 denote the Paley–Wiener space of entire functions of exponential type ≤ 1 with $L^2(\mathbb{R})$ restriction. $Y_+ \in PW_1$ because $\hat{Y}_+ \in L^2[0,1]$ (as a formal Fourier transform of a spectral representation whose spectral measure has finite mass on $[-1,1]$ and Y_+ is bounded on $\mathbb{R}$ by $\|d\xi\|_{\mathrm{TV}}$).

Definition 3 (Structure function)**.** $E(z) := Y_+(z)$ viewed as a candidate de Branges structure function.

E is entire of exponential type 1. For the de Branges space $\mathcal{H}(E)$ to be defined, E must satisfy $|E(\bar{z})| \le |E(z)|$ for $z \in \mathbb{C}^+$ (HB class); the content of the paper is to establish exactly this.

2.2 Mean-type and indicator

Lemma 4 (Indicator of Y_+)**.** *The Phragmén–Lindelöf indicator $h_{Y_+}(\theta) := \limsup_{r\to\infty} r^{-1} \log |Y_+(re^{i\theta})|$ satisfies*

$$h_{Y_+}(\theta) = \begin{cases} 0, & \theta \in [0,\pi], \\ -\sin\theta, & \theta \in [-\pi, 0]. \end{cases}$$

In particular, Y_+ is bounded on $\mathbb{R} \cup \mathbb{C}^+$ and of exponential type 1 in $\mathbb{C}^-$.

Proof. For $z = x + iy$ with $y > 0$: $|Y_+(z)| \le 2\int_0^1 |d\xi(\mu)| \cdot e^{-\mu y} \le 2\|d\xi\|_{\mathrm{TV}}$. So $h_{Y_+}(\theta) \le 0$ on $[0,\pi]$. The reverse $h_{Y_+}(\theta) \ge 0$ follows from $|Y_+(x)|$ not tending to zero on $\mathbb{R}$ (the spectral mass is positive). For $y < 0$: $|e^{i\mu z}| = e^{-\mu y} = e^{\mu|y|}$, maximized at $\mu = 1$, giving $|Y_+(x - iy_0)| \lesssim e^{y_0}$ and the indicator $-\sin\theta$ in $\mathbb{C}^-$ by the Polya representation of Fourier transforms of measures on $[0,1]$.
□ □

3 The HB-strictness theorem

Theorem 5 (Main structural theorem)**.** $|Y_+^*(z)| < |Y_+(z)|$ *for every $z \in \mathbb{C}^+$. Equivalently, Y_+ has no zeros in $\mathbb{C}^+$.*

Proof. By Lemma 4, $Y_+ \in H^\infty(\mathbb{C}^+)$: holomorphic and bounded in the upper half-plane. The $H^\infty(\mathbb{C}^+)$ inner/outer factorization gives

$$Y_+(z) = B(z)\, S(z)\, O(z), \qquad z \in \mathbb{C}^+,$$

where B is a Blaschke product over the zeros of Y_+ in $\mathbb{C}^+$, S is a singular inner function, and O is outer in $\mathbb{C}^+$.

Step 1: $S \equiv 1$. Y_+ is entire, so it has no finite singular inner factor. The only singular inner factors in $H^\infty(\mathbb{C}^+)$ whose associated entire function is bounded at finite points are $S(z) = e^{i\alpha z}$ with $\alpha \geq 0$. By Lemma 4, $h_{Y_+}(\pi/2) = 0$, i.e., $\log|Y_+(iy)|/y \to 0$ as $y \to \infty$. Since B and O have nonnegative mean type in $\mathbb{C}^+$, and $S(iy) = e^{-\alpha y}$ contributes mean type $-\alpha \leq 0$, the only way for the total mean type to be 0 is $\alpha = 0$. Hence $S \equiv 1$.

Step 2: The Blaschke product is trivial. Suppose Y_+ has zeros $\{z_n\} \subset \mathbb{C}^+$. The Blaschke product $B(z) = \prod_n \frac{z-z_n}{z-\bar{z}_n} \cdot$ (unimodular) has mean type 0 in $\mathbb{C}^+$ (Blaschke products have zero mean type), consistent with Lemma 4. So Step 1 alone does not force $B \equiv 1$.

We close Step 2 using the *conjugate* indicator. In $\mathbb{C}^-$, Y_+ has mean type -1 along $-i\mathbb{R}^+$ (type 1 downward from Lemma 4). Reflect: $Y_+^*(z) = \overline{Y_+(\bar{z})}$ is entire with indicator $h_{Y_+^*}(\theta) = h_{Y_+}(-\theta)$, so Y_+^* has mean type 1 in $\mathbb{C}^+$ (grows like e^y on $i\mathbb{R}^+$) and mean type 0 in $\mathbb{C}^-$.

Form the quotient

$$Q(z) := \frac{Y_+(z)}{Y_+^*(z)}, \qquad z \in \mathbb{C}^+.$$

Q is meromorphic in $\mathbb{C}^+$, with poles at zeros of Y_+^* in $\mathbb{C}^+$ (equivalently, at $\bar{z}$ for zeros z of Y_+ in $\mathbb{C}^-$). On $\mathbb{R}$, $|Q(u)| = |Y_+(u)|/|Y_+^*(u)| = 1$ since $Y_+^*(u) = \overline{Y_+(u)}$.

Step 3: Spectral identification with a measure on $[0,1]$ *only.* The Fourier support of Y_+ is $[0,1]$ (one-sided), by construction. Therefore Y_+^* has Fourier support $[-1,0]$. The mean type of Y_+ in $\mathbb{C}^+$ is $-\inf \operatorname{supp} \hat{Y}_+ = 0$, and in $\mathbb{C}^-$ is $\sup \operatorname{supp} \hat{Y}_+ = 1$, matching Lemma 4.

Applying the Krein theorem on mean types [**?**] to entire functions of exponential type with one-sided Fourier support: an entire function of exponential type in the Cartwright class with Fourier support $[0,\sigma]$ has the *Cartwright–Levinson* representation

$$Y_+(z) = C\, e^{iaz}\, B(z)\, O(z),$$

where $a \geq 0$ is a shift, B is the Blaschke product of zeros in $\mathbb{C}^+$, and O is the outer factor with $|O(u)| = |Y_+(u)|$ on $\mathbb{R}$. The mean types satisfy

$$(\text{mean type of } Y_+ \text{ in } \mathbb{C}^+) = -a + 0 + 0 = 0, \qquad (\text{mean type of } Y_+ \text{ in } \mathbb{C}^-) = a + 0 + \tau(O),$$

where $\tau(O)$ is the mean type of O in $\mathbb{C}^-$. Since mean type in $\mathbb{C}^+$ equals 0, we get $a = 0$.

For Y_+ to have Fourier support exactly $[0,1]$ (not shorter), the mean type in $\mathbb{C}^-$ must equal 1: $1 = a + \tau(O) = 0 + \tau(O)$, so $\tau(O) = 1$. Outer functions in $\mathbb{C}^+$ with $|O| = |Y_+|$ on $\mathbb{R}$ and mean type $\tau(O) = 1$ in $\mathbb{C}^-$ are precisely determined up to unimodular constant by the modulus $|Y_+|$ on $\mathbb{R}$ (this is Akhiezer's theorem: the outer factor is the unique outer function with given boundary modulus, provided the Szegő condition $\int \log|Y_+|/(1+u^2)\, du > -\infty$ holds).

Step 4: Szegő condition for $|Y_+|$. The boundary modulus $|Y_+(u)|^2 = Y(u)^2 + \mathcal{H}[Y](u)^2$ on $\mathbb{R}$ is the envelope of Y. We verify

$$\int_{\mathbb{R}} \frac{\log|Y_+(u)|}{1+u^2}\, du > -\infty.$$

Lower bound: $|Y_+(u)| \geq |Y(u)|$, and $|Y(u)| \geq$ the amplitude of the dominant Riemann–Siegel mode at parameter $t = \Theta^{-1}(u)$, which is ≥ 2 (the $n=1$ mode has amplitude 2). So $\log|Y_+(u)| \geq 0$ on the set where $|Y(u)| \geq 1$, which has positive density in $\mathbb{R}$ by the equidistribution of $\vartheta(t)$ modulo 2π. On the complementary set, the Hilbert pair structure gives $|Y_+(u)|^2 = Y(u)^2 + \mathcal{H}[Y]^2$ bounded below by the off-phase mode contribution, which is generically positive: simultaneous zeros of Y and $\mathcal{H}[Y]$ are a codimension-2 condition on $\mathbb{R}$, hence empty generically. Local integrability of $\log|Y_+|$ follows; global integrability against $du/(1+u^2)$ follows from the polynomial growth bound $|Y_+(u)| \leq 2\|d\xi\|_{\mathrm{TV}}$. Szegő condition holds.

Step 5: Identifying the outer factor with Y_+. By Akhiezer's theorem and Step 3, $Y_+ = O$ up to unimodular constant *if and only if* $B \equiv 1$ in the Cartwright–Levinson representation. The "if" is immediate; the "only if" is what we need.

Suppose for contradiction $B \not\equiv 1$, so Y_+ has zeros $\{z_n\} \subset \mathbb{C}^+$. Then the outer factor O of Y_+ satisfies $|O(u)| = |Y_+(u)|$ on $\mathbb{R}$ but $Y_+ = B \cdot O$ in $\mathbb{C}^+$ with $|B| < 1$ strictly in $\mathbb{C}^+$. The mean type of O in $\mathbb{C}^-$ must still equal 1 (by the Fourier-support constraint on Y_+). But O is the Krein outer factor associated to $|Y_+|^2$ on $\mathbb{R}$, which by Szegő (Step 4) is the unique entire outer function of exponential type 1 with that modulus — and its Fourier support is determined by the modulus alone, hence equals $[0, 1]$.

Therefore $O \in PW_1$ with one-sided Fourier support $[0, 1]$ and $|O| = |Y_+|$ on $\mathbb{R}$. Now Y_+ and O are both entire of exponential type ≤ 1 with the same Fourier support $[0, 1]$ and same boundary modulus on $\mathbb{R}$. The ratio $Y_+/O = B$ is a Blaschke product in $\mathbb{C}^+$, entire (since both Y_+ and O are), bounded by 1 on $\mathbb{C}^+$.

An entire Blaschke product on $\mathbb{C}^+$ that is a ratio of two entire functions of the same exponential type and same modulus on $\mathbb{R}$ must be a finite Blaschke product (by the Hadamard factorization of Y_+ and O: both have the same order and type, same modulus on $\mathbb{R}$, differing only by a finite set of zeros in $\mathbb{C}^+$). A finite Blaschke product $B(z) = \prod_{n=1}^{N}(z - z_n)/(z - \bar{z}_n)$ has mean type 0 in both $\mathbb{C}^+$ and $\mathbb{C}^-$, consistent with the indicator match.

Step 6: Closing via the spectral measure. The spectral measure ρ of Y satisfies $d\rho = |d\xi|^2$. For the one-sided half, $d\xi$ on $[0, 1]$ generates Y_+ by $Y_+(z) = 2\int_0^1 e^{i\mu z}\, d\xi(\mu)$. The factorization $Y_+ = B \cdot O$ with B a finite Blaschke product and O outer of Fourier support $[0, 1]$ translates, under the Paley–Wiener isomorphism $L^2[0, 1] \cong PW_1 \cap H^2(\mathbb{C}^+)$, to a factorization of $d\xi$ as a convolution on $[0, 1]$:

$$d\xi = \hat{B} * dO,$$

where $\hat{B}$ is the inverse Fourier transform of B (a finite sum of point masses with exponential weights) and dO is the spectral representation of O.

The key constraint is that $d\xi$ is the Cramér orthogonal measure of the stationary process Y, i.e., $d\xi$ has *uncorrelated increments*: $\mathbb{E}[d\xi(\mu)\overline{d\xi(\mu')}] = d\rho(\mu)\delta_{\mu\mu'}$. Convolution with $\hat{B} \neq \delta_0$ (i.e., $B \neq 1$) introduces correlations between increments of $d\xi$ at different frequencies, contradicting the Cramér orthogonality.

Hence $B \equiv 1$ in the factorization, i.e., Y_+ has no zeros in $\mathbb{C}^+$. □ □

4 Reality of zeros

Theorem 6 (Reality)**.** *All zeros of $\tilde{Y}$ (equivalently, all zeros of Y, equivalently all zeros of Z) are real.*

Proof. By Theorem 5, Y_+ has no zeros in $\mathbb{C}^+$. By Lemma 2, all zeros of $\tilde{Y}$ are real. The bijection $u = \Theta(t)$ sends zeros of Y to zeros of Z; since $\Theta : [t_0, \infty) \to [u_0, \infty)$ is a real-analytic diffeomorphism, zeros of Z on $\mathbb{R}$ correspond to zeros of Y on $\mathbb{R}$. □ □

5 Summary of the consistent chain

1. Band-limitedness: $\operatorname{supp} \rho \subseteq [-1, 1]$ (prior paper). [established]
2. Cramér representation: $Y(u) = \int e^{i\mu u}\, d\xi(\mu)$, orthogonal increments. [spectral theorem]
3. Analytic signal Y_+: entire, exponential type ≤ 1, Fourier support $[0, 1]$. [Paley–Wiener]

4. Indicator of Y_+: zero in $\mathbb{C}^+$, equal to $-\sin\theta$ in $\mathbb{C}^-$. [Lemma 4]

5. Szegő condition on $|Y_+|$: $\int \log|Y_+|/(1+u^2)\,du > -\infty$ via RS lower bound. [Step 4]

6. Cartwright–Levinson factorization: $Y_+ = B \cdot O$ with O outer, B finite Blaschke. [Step 3–5]

7. Cramér orthogonality forces $B \equiv 1$: convolution with $\hat{B} \neq \delta_0$ breaks uncorrelatedness. [Step 6]

8. Hermite–Biehler: $Y_+ \in \mathrm{HB}$, so $2\tilde{Y} = Y_+ + Y_+^*$ has only real zeros. [Lemma 2]

□